Scientific Computing 2026

Entropy, KL divergence, and Fisher information

Lecture 7

A rare outcome carries more surprise

Assume a population with two labels:

$$\mathbb{P}(Y = 1) = 0.8, \qquad \mathbb{P}(Y = 0) = 0.2.$$

Definition: Surprisal

An outcome of probability $p > 0$ has surprisal $-\log p$. We use natural logarithms; the unit is a *nat*.

$$-\log 0.8 \approx 0.2231, \qquad -\log 0.2 \approx 1.6094.$$

For independent outcomes, surprise adds:

$$-\log(p_1 p_2) = -\log p_1 - \log p_2.$$

Entropy averages the surprise

Let $P = (p_1, \ldots, p_K)$ be probabilities for K labels.

Definition: Entropy

$$H(P) = \mathbb{E}_{Y \sim P}[-\log p_Y] = -\sum_{k=1}^{K} p_k \log p_k, \qquad 0 \log 0 := 0.$$

For the population with 80% successes,

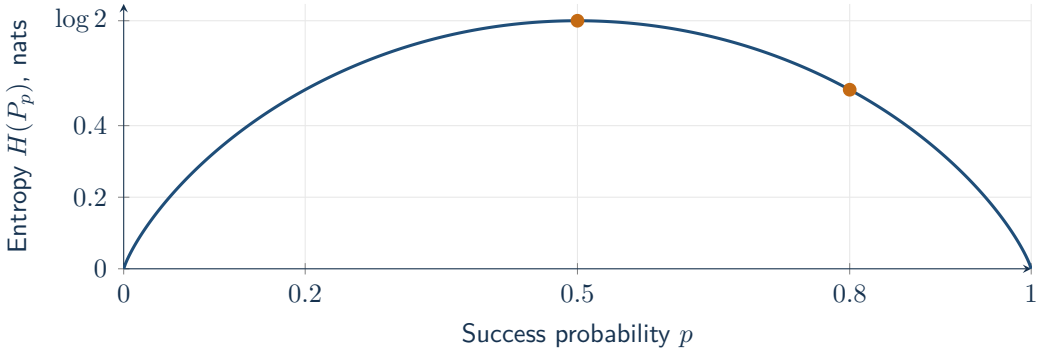
$$H(P) = -0.8 \log 0.8 - 0.2 \log 0.2 \approx 0.5004.$$

If one label has probability one, $H(P) = 0$.

Bernoulli entropy is largest at $p = 1/2$

Write $P_p = \text{Bernoulli}(p)$:

$$H(P_p) = -p \log p - (1-p) \log(1-p).$$



A balanced label is hardest to predict. The maximum is $\log 2 \approx 0.6931$ nats.

Evaluate predictions using the actual outcomes

$P = (p_1, \ldots, p_K)$: population probabilities.
$Q = (q_1, \ldots, q_K)$: predicted probabilities for the same labels.

Definition: Cross-entropy

$$\mathrm{CE}(P, Q) = \mathbb{E}_{Y \sim P}[-\log q_Y] = -\sum_{k=1}^{K} p_k \log q_k.$$

If the population success probability is 0.8, but we predict q,

$$\mathrm{CE}(P_{0.8}, P_q) = -0.8 \log q - 0.2 \log(1 - q).$$

If $p_k > 0$ but $q_k = 0$, the expected loss is infinite.

Overconfident errors are expensive

Keep the population success probability fixed at 0.8.

Predicted q	Expected log loss	Excess over 0.5004
0.50	0.6931	0.1927
0.60	0.5919	0.0915
0.80	0.5004	0
0.99	0.9291	0.4287

At $q = 0.99$, a failure has loss $-\log 0.01 \approx 4.6052$. Failures occur 20% of the time in this population.

KL is the extra expected log loss

Definition: Kullback–Leibler divergence

$$D_{\mathrm{KL}}(P\|Q) = \sum_{k:p_k>0} p_k \log \frac{p_k}{q_k}.$$

It is infinite if a positive p_k is paired with $q_k = 0$.

Subtract the entropy from the cross-entropy:

$$-\sum_k p_k \log q_k + \sum_k p_k \log p_k = \sum_k p_k \log \frac{p_k}{q_k}.$$

Result: Entropy plus mismatch

$$\mathrm{CE}(P, Q) = H(P) + D_{\mathrm{KL}}(P\|Q).$$

Why KL cannot be negative

Use the scalar inequality $-\log u \geq 1 - u$, with equality only at $u = 1$. For positive p_k, set $u = q_k/p_k$:

$$\begin{aligned} D_{\mathrm{KL}}(P\|Q) &= \sum_{k:p_k>0} p_k \left(-\log \frac{q_k}{p_k}\right) \\ &\geq \sum_{k:p_k>0} (p_k - q_k) \geq 0. \end{aligned}$$

Result: Best prediction for a fixed population

$D_{\mathrm{KL}}(P\|Q) \geq 0$, with equality exactly when $Q = P$.
Therefore $\mathrm{CE}(P, Q) \geq H(P)$.

The direction of KL changes the answer

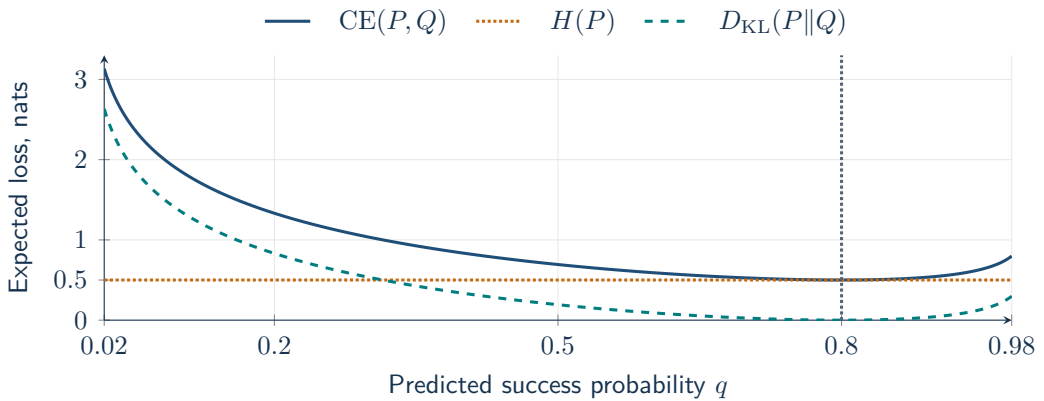
For $P = P_{0.8}$ and $Q = P_{0.5}$,

$$D_{\mathrm{KL}}(P\|Q) = 0.8 \log \frac{0.8}{0.5} + 0.2 \log \frac{0.2}{0.5} \approx 0.1927,$$

$$D_{\mathrm{KL}}(Q\|P) = 0.5 \log \frac{0.5}{0.8} + 0.5 \log \frac{0.5}{0.2} \approx 0.2231.$$

The first distribution supplies the averaging probabilities. KL is not symmetric, so it is not a distance.

Entropy stays fixed; mismatch changes



Population $p = 0.8$. The vertical gap between cross-entropy and entropy is KL. The best prediction is $q = 0.8$.

Maximum likelihood minimizes empirical cross-entropy

For n independent labels, let $\widehat{p}_k = n_k/n$, where n_k counts label k. The model predicts $q_\theta(k)$.

$$-\frac{1}{n}\log L(\theta) = -\frac{1}{n}\sum_{i=1}^{n}\log q_\theta(y_i)$$
$$= -\sum_{k=1}^{K}\widehat{p}_k \log q_\theta(k) = \mathrm{CE}(\widehat{P}, Q_\theta).$$

Result: Same minimizer

$$\mathrm{CE}(\widehat{P}, Q_\theta) = \underbrace{H(\widehat{P})}_{\text{fixed data}} + D_{\mathrm{KL}}(\widehat{P}\|Q_\theta).$$

This empirical KL identity is for finite labels, not continuous densities.

Multiclass loss selects the observed label

For input x, the model predicts $q_\theta(k \mid x)$. If the observed label is y, the one-hot target is

$$t_k = \begin{cases} 1, & k = y, \\ 0, & k \neq y. \end{cases}$$

Its cross-entropy is

$$-\sum_{k=1}^{K} t_k \log q_\theta(k \mid x) = -\log q_\theta(y \mid x).$$

Example: $y = 2$ and predicted probabilities $(0.6, 0.3, 0.1)$:

$$\text{loss} = -\log 0.3 \approx 1.2040.$$

Language-model loss is conditional log loss

For an observed sequence $w_1, \ldots, w_m$, the chain rule gives

$$Q_\theta(w_1, \ldots, w_m) = \prod_{t=1}^{m} q_\theta(w_t \mid w_1, \ldots, w_{t-1}).$$

The condition is empty for $t = 1$. The average token loss is

$$-\frac{1}{m} \sum_{t=1}^{m} \log q_\theta(w_t \mid w_1, \ldots, w_{t-1}).$$

If three observed tokens receive probabilities $0.5,\ 0.25,\ 0.1$,

$$Q_\theta(\text{sequence}) = 0.0125, \qquad \text{average loss} = -\frac{\log 0.0125}{3} \approx 1.4607.$$

The tokens do not need to be independent.

Distillation uses probability targets

For the same input, a teacher gives probabilities P; a student gives probabilities Q:

$$P = (0.7, 0.2, 0.1), \qquad Q = (0.6, 0.25, 0.15).$$

Train the student using all teacher probabilities:

$$\mathrm{CE}(P, Q) = -0.7 \log 0.6 - 0.2 \log 0.25 - 0.1 \log 0.15 \approx 0.8245.$$

$$\underbrace{\mathrm{CE}(P, Q)}_{0.8245} = \underbrace{H(P)}_{0.8018} + \underbrace{D_{\mathrm{KL}}(P \| Q)}_{0.0227}.$$

The teacher entropy is fixed, so minimizing cross-entropy minimizes this KL.

A KL penalty can keep predictions near a reference

For a specified input, let Q_θ be the model prediction and Q_{ref} a fixed reference on the same labels.

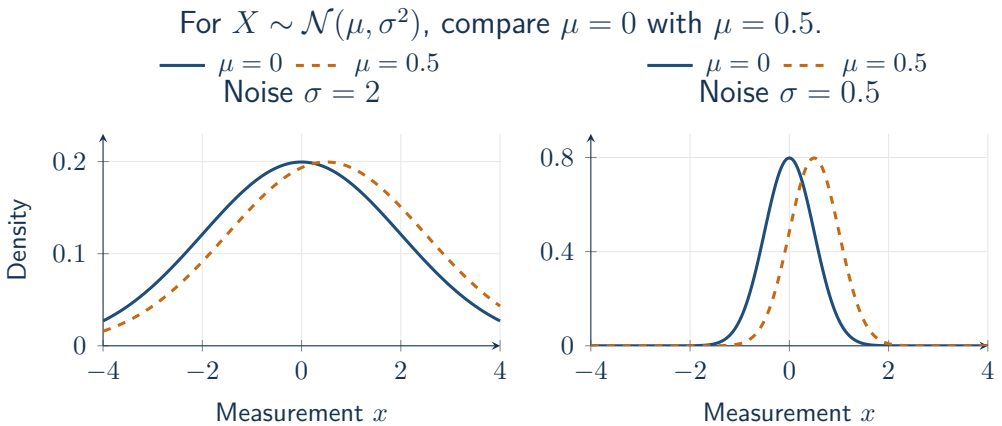
$$\mathcal{L}_{\text{train}}(\theta) + \beta D_{\text{KL}}(Q_\theta \| Q_{\text{ref}}), \qquad \beta \geq 0.$$

$\mathcal{L}_{\text{train}}$ is the training loss; β controls the penalty strength.

$$D_{\text{KL}}(Q_\theta \| Q_{\text{ref}}) = \sum_k q_\theta(k) \log \frac{q_\theta(k)}{q_{\text{ref}}(k)}.$$

A reference probability of zero forbids positive model mass there when $\beta > 0$. This penalty concerns outputs; MAP used a prior on parameters.

Distinguishing two possible means



Both panels use the same horizontal scale and a mean shift of 0.5. Less noise makes the two possible means easier to distinguish from data.

The score tells us which way an observation pushes

Let $f_\theta(x)$ be a PMF or density with scalar parameter θ.

Definition: Score of one observation

$$s_\theta(x) = \frac{\partial}{\partial \theta} \log f_\theta(x).$$

For $X \sim \mathcal{N}(\mu, \sigma^2)$, with known σ^2,

$$s_\mu(x) = \frac{x - \mu}{\sigma^2}.$$

If $x > \mu$, increasing μ raises this observation's log-likelihood. If $x < \mu$, decreasing μ raises it.

Observed information at the fitted maximum

For the recorded sample, let $\ell_n(\theta) = \log L(\theta)$. Its derivative is the sample score $S_n(\theta) = \ell_n'(\theta)$.

$$S_n(\widehat{\theta}) = 0 \qquad \text{at a differentiable interior MLE.}$$

Both a flat peak and a sharp peak have zero slope.

Definition: Observed information

$$J_n(\theta) = -\ell_n''(\theta).$$

It describes the downward curvature for this particular dataset.

At a fitted peak, larger positive J_n means a sharper peak.

The likelihood drop near its maximum

Let δ be a small change from the MLE $\widehat{\theta}$. The zero slope leaves the quadratic term:

$$\ell_n(\widehat{\theta}+\delta) - \ell_n(\widehat{\theta}) \approx -\frac{1}{2} J_n(\widehat{\theta})\delta^2.$$

For the same change $\delta = 0.1$:

Curvature $J_n(\widehat{\theta})$	Log-likelihood change	Likelihood ratio
4	-0.02	$e^{-0.02} \approx 0.980$
100	-0.5	$e^{-0.5} \approx 0.607$

The ratio compares $L(\widehat{\theta}+0.1)$ with $L(\widehat{\theta})$. The larger curvature gives a larger drop.

Fisher information averages over possible datasets

Keep θ fixed. Draw new datasets from that model and calculate their curvatures at the same θ.

Definition: Fisher information

For a regular model,

$$I_n(\theta) = \mathbb{E}_\theta[J_n(\theta)].$$

For one observation,

$$I_1(\theta) = -\mathbb{E}_\theta[\partial_\theta^2 \log f_\theta(X)].$$

J_n depends on the observed sample. I_n averages over possible samples, without refitting θ each time.

Why the score has mean zero

Assume common support, an interior parameter, and differentiability allowing derivatives through the sum or integral.

$$\mathbb{E}_\theta[s_\theta(X)] = \int \frac{\partial_\theta f_\theta(x)}{f_\theta(x)} f_\theta(x)\, dx$$
$$= \frac{\partial}{\partial\theta} \int f_\theta(x)\, dx = \frac{\partial}{\partial\theta} 1 = 0.$$

For a discrete law, replace the integral by a sum. For the Gaussian mean:

$\mathbb{E}_\mu[(X-\mu)/\sigma^2] = 0.$

Fisher information as an expected squared score

Differentiate the log-likelihood twice:

$$\partial_\theta^2 \log f_\theta = \frac{\partial_\theta^2 f_\theta}{f_\theta} - \left(\frac{\partial_\theta f_\theta}{f_\theta}\right)^2.$$

The first term has expectation $\int \partial_\theta^2 f_\theta(x)\, dx = \partial_\theta^2 1 = 0$.

Result: Equivalent score formula

With a finite score second moment,

$$I_1(\theta) = \mathbb{E}_\theta[s_\theta(X)^2] = \mathrm{Var}_\theta(s_\theta(X)).$$

The fitted sample score is zero. Scores on new datasets at a fixed parameter vary, so their expected square need not be zero.

Gaussian noise and information about the mean

For the Gaussian mean with known variance,

$$I_1(\mu) = \frac{\mathbb{E}_\mu[(X-\mu)^2]}{\sigma^4} = \frac{1}{\sigma^2}.$$

Observation noise	σ	Information $I_1(\mu)$
More noise	2	0.25
Less noise	0.5	4

These are the two noise levels in the earlier density plots. The smaller noise gives sixteen times as much information about μ.

Independent observations add information

For n i.i.d. observations,

$$\ell_n(\theta) = \sum_{i=1}^{n} \log f_\theta(X_i), \qquad S_n(\theta) = \ell_n'(\theta) = \sum_{i=1}^{n} s_\theta(X_i).$$

Independence and zero mean scores give

$$I_n(\theta) = \mathrm{Var}_\theta(S_n(\theta)) = \sum_{i=1}^{n} \mathrm{Var}_\theta(s_\theta(X_i)) = nI_1(\theta).$$

For Gaussian measurements with known variance σ^2,

$$I_n(\mu) = \frac{n}{\sigma^2}.$$

Gaussian mean: observed and expected information

For n independent measurements with known variance σ^2,

$$\ell_n(\mu) = C - \frac{n}{2\sigma^2}(\mu - \bar{x})^2,$$

where C does not depend on the candidate mean μ.

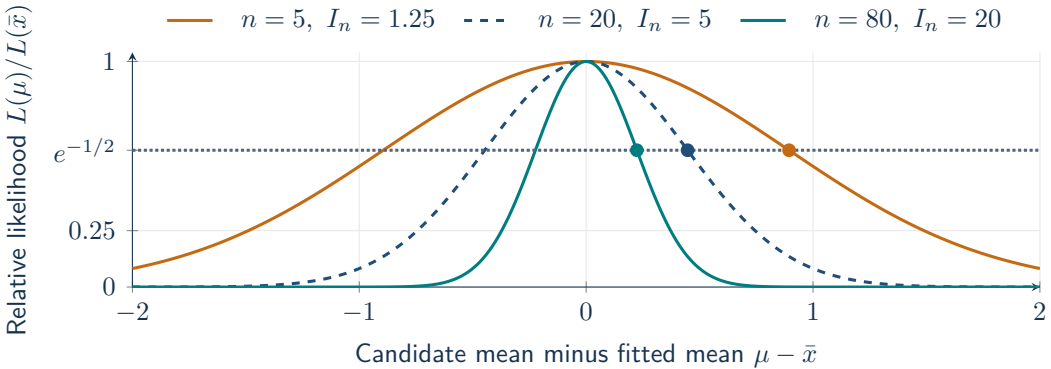
$$J_n(\mu) = -\ell_n''(\mu) = \frac{n}{\sigma^2} = I_n(\mu).$$

Result: The same curvature for every possible dataset

$$\frac{L(\mu)}{L(\bar{x})} = \exp\left[-\frac{1}{2} I_n(\mu)(\mu - \bar{x})^2\right].$$

Larger information gives a narrower likelihood peak

Known $\sigma = 2$, so $J_n = I_n = n/4$. At a distance $1/\sqrt{I_n}$ from the peak, relative likelihood is $e^{-1/2}$.



Marked distances: $0.894,\ 0.447,\ 0.224$. Larger I_n gives a narrower peak.

Bernoulli score: successes and failures pull differently

Now the success probability $p \in (0,1)$ is unknown. For $Y \sim \text{Bernoulli}(p)$,

$$\log f_p(y) = y \log p + (1-y)\log(1-p),$$

$$s_p(y) = \frac{y}{p} - \frac{1-y}{1-p} = \frac{y-p}{p(1-p)}.$$

At $p = 0.8$,

$$s_p(1) = 1.25, \qquad s_p(0) = -5, \qquad 0.8(1.25) + 0.2(-5) = 0.$$

A rare failure gives a larger push toward lowering p.

Bernoulli Fisher information

Use $\mathrm{Var}(Y) = p(1-p)$:

$$I_1(p) = \mathbb{E}_p\left[\frac{(Y-p)^2}{p^2(1-p)^2}\right] = \frac{p(1-p)}{p^2(1-p)^2} = \frac{1}{p(1-p)}.$$

Result: A sample of independent labels

$$I_n(p) = \frac{n}{p(1-p)}.$$

At $p = 0.8$, one label gives $I_1 = 6.25$; 100 labels give $I_{100} = 625$.

Observed and expected information can differ

For 10 Bernoulli trials, let k be the observed success count:

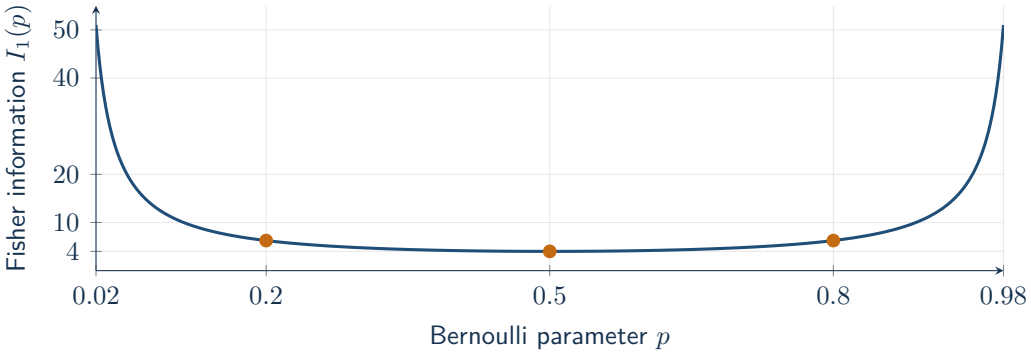
$$J_{10}(p) = \frac{k}{p^2} + \frac{10-k}{(1-p)^2}.$$

Evaluate different datasets at the same fixed $p = 0.8$:

Successes k	Observed $J_{10}(0.8)$	Expected $I_{10}(0.8)$
7	85.94	62.5
9	39.06	62.5

Fisher information averages over all possible success counts, using their probabilities under $p = 0.8$.

Bernoulli entropy and information



Entropy is largest at $p = 1/2$; information about p is smallest there. The sample proportion's absolute variance, $p(1-p)/n$, is also largest at $p = 1/2$.

How small can an unbiased estimator's variance be?

Result: Cramér–Rao lower bound

For a regular model, let T estimate the scalar parameter θ. If T has finite variance, is unbiased throughout an open parameter interval, and differentiation of its expectation is valid, then

$$\operatorname{Var}_\theta(T) \geq \frac{1}{I_n(\theta)} = \frac{1}{nI_1(\theta)}, \qquad 0 < I_n(\theta) < \infty.$$

Definition: Efficient unbiased estimator

An unbiased estimator attaining this bound is efficient at that parameter value.

Cramér–Rao proof: differentiating unbiasedness

Write $x = (x_1, \ldots, x_n)$, and let $g_\theta(x)$ be the joint sample density. The rule $T(x)$ does not use the unknown θ.

$$\mathbb{E}_\theta T = \theta \quad \Longrightarrow \quad 1 = \frac{\partial}{\partial \theta} \int T(x) g_\theta(x)\, dx$$
$$= \int T(x)\, \partial_\theta g_\theta(x)\, dx.$$

Use $\partial_\theta g_\theta = g_\theta\, \partial_\theta \log g_\theta$. The log-derivative is the sample score $S_n(\theta)$, hence

$$1 = \int T(x) g_\theta(x)\, \partial_\theta \log g_\theta(x)\, dx = \mathbb{E}_\theta[T S_n(\theta)].$$

For a discrete sample, use sums instead of integrals.

Cramér–Rao proof: the covariance bound

The previous step gave $\mathbb{E}_\theta[T S_n(\theta)] = 1$. Since $\mathbb{E}_\theta S_n(\theta) = 0$,

$$\begin{aligned}\operatorname{Cov}_\theta(T, S_n(\theta)) &= \mathbb{E}_\theta[T S_n(\theta)] - \mathbb{E}_\theta T \, \mathbb{E}_\theta S_n(\theta) \\ &= 1 - \theta \cdot 0 = 1.\end{aligned}$$

Lecture 3's inequality $\operatorname{Cov}(U,V)^2 \le \operatorname{Var}(U)\operatorname{Var}(V)$ now gives

$$1 \le \operatorname{Var}_\theta(T) \underbrace{\operatorname{Var}_\theta(S_n(\theta))}_{I_n(\theta)}.$$

Divide by $I_n(\theta) > 0$: $\operatorname{Var}_\theta(T) \ge 1/I_n(\theta)$.

The Gaussian sample mean attains the bound

For $X_i \sim \mathcal{N}(\mu, \sigma^2)$ independently, with known σ^2,

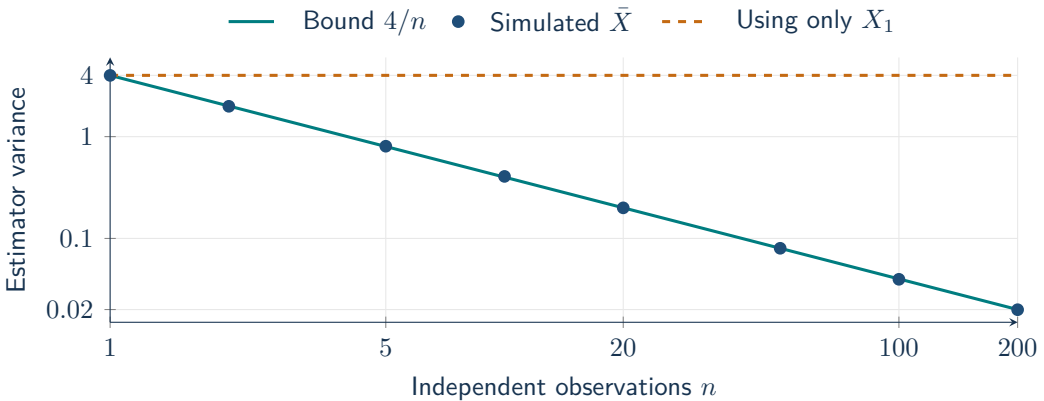
$$I_n(\mu) = \frac{n}{\sigma^2}, \qquad \mathrm{Var}_\mu(\bar{X}_n) = \frac{\sigma^2}{n} = \frac{1}{I_n(\mu)}.$$

The sample mean is unbiased and efficient.

Estimator	Expectation	Variance
$\bar{X}_n$	μ	σ^2/n
X_1	μ	σ^2

Using only the first measurement is unbiased but discards data.

The sample mean follows the information bound



$X_i \sim \mathcal{N}(\mu, 4)$ independently. The points use $30{,}000$ simulated datasets per n. The first-observation estimator has variance 4 at every n.

The Bernoulli sample proportion is also efficient

For n independent Bernoulli labels, with $0 < p < 1$,

$$\widehat{p} = \frac{1}{n}\sum_{i=1}^{n} Y_i, \qquad \mathbb{E}_p \widehat{p} = p.$$

$$\mathrm{Var}_p(\widehat{p}) = \frac{p(1-p)}{n} = \frac{1}{I_n(p)}.$$

At $p = 0.8,\ n = 100$,

$$\mathrm{Var}_p(\widehat{p}) = 0.0016, \qquad \mathrm{SE}_p(\widehat{p}) = 0.04.$$

This variance is exact. A Gaussian approximation to the sampling distribution is still an approximation.

A biased estimator can have smaller variance

For the Gaussian mean, consider $T = \bar{X}_n/2$:

$$\mathrm{Var}_\mu(T) = \frac{\sigma^2}{4n} < \underbrace{\frac{\sigma^2}{n}}_{\text{unbiased variance bound}}, \qquad \mathbb{E}_\mu T - \mu = -\frac{\mu}{2}.$$

Its mean squared error includes both terms:

$$\mathbb{E}_\mu[(T-\mu)^2] = \frac{\sigma^2}{4n} + \frac{\mu^2}{4}.$$

The Cramér–Rao statement above requires unbiasedness.

A Gaussian approximation for an MLE

The Bernoulli MLE is a sample average, so its Gaussian approximation comes from the CLT. Regular MLEs have a similar large-sample behavior.

Result: Approximate sampling distribution

For true parameter θ and MLE $\widehat{\theta}$,

$$\widehat{\theta} \text{ is approximately } \mathcal{N}\left(\theta, \frac{1}{I_n(\theta)}\right).$$

This requires a consistent MLE away from parameter boundaries and regular likelihood behavior near the true value. It gives no finite-sample unbiasedness guarantee.

Estimating the error scale from eighty successes

The probability p is unknown. Observing 80 successes in 100 independent trials gives $\widehat{p} = 0.8$.

$$I_{100}(p) = \frac{100}{p(1-p)} \qquad \text{so} \qquad I_{100}(\widehat{p}) = \frac{100}{0.8(0.2)} = 625.$$

Result: Estimated standard error

$$\widehat{\mathrm{SE}}(\widehat{p}) = \frac{1}{\sqrt{I_{100}(\widehat{p})}} = \frac{1}{\sqrt{625}} = 0.04.$$

We substitute the fitted probability for the unknown p to estimate the sampling spread.

The corresponding confidence interval

For the same 100 trials, $\widehat{p} = 0.8$ and $\widehat{\mathrm{SE}} = 0.04$.

Result: Approximate 95% confidence interval

$$\widehat{p} \pm 1.96\,\widehat{\mathrm{SE}} = 0.8 \pm 0.0784 = [0.7216,\ 0.8784].$$

This is the Wald interval from Lecture 6. Fisher information gives its error scale. The Gaussian approximation can still be poor near $p = 0$ or $p = 1$, especially with few observations.

Nearby distributions and Fisher information

Let P_θ be a regular model and let δ be a small change in its parameter.

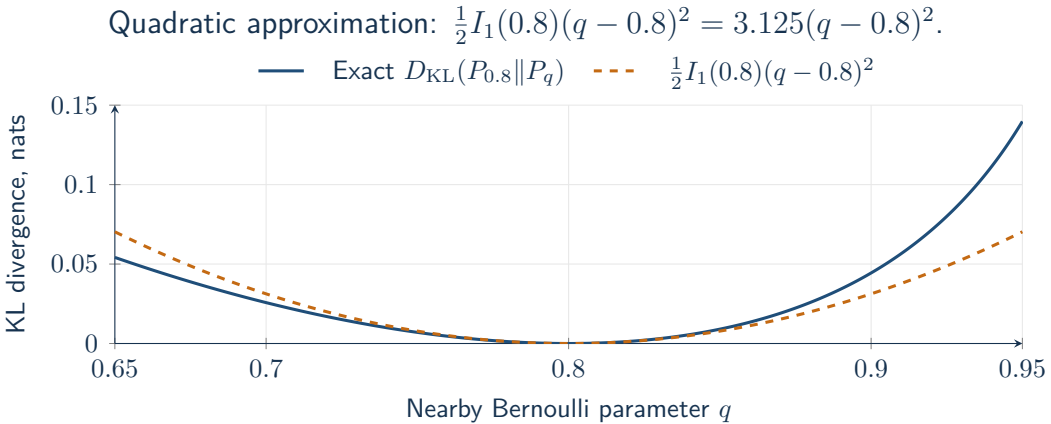
Result: KL for a small parameter change

$$D_{\mathrm{KL}}(P_\theta \| P_{\theta+\delta}) \approx \frac{1}{2} I_1(\theta) \delta^2.$$

KL is zero when the two parameter values coincide. Fisher information sets how quickly it grows for small changes. For Bernoulli $p = 0.8$, $I_1(p) = 6.25$, giving

$$D_{\mathrm{KL}}(P_{0.8} \| P_{0.8+\delta}) \approx 3.125\, \delta^2.$$

Bernoulli KL near $p = 0.8$



Here $\delta = q - 0.8$. The quadratic curve agrees near $q = 0.8$, then loses accuracy farther away.